

Yushin Choi

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RESEARCH INTERESTS

Adaptive humanoid loco-manipulation through whole-body control and learning-based motion generation, with a focus on generative motion models

EDUCATION

Hanyang University

Master of Science in Intelligence and Information Engineering

Seoul, Republic of Korea

Mar. 2023 – Feb. 2025

- Advisor: Prof. Wansoo Kim
- Thesis: Hierarchical Whole-Body Control for Enhanced Adaptability in Mobile Manipulators
- GPA: 4.33/4.5

Hanyang University ERICA

Bachelor of Science in Robotics, summa cum laude (early graduation)

Ansan, Republic of Korea

Mar. 2017 – Feb. 2023

- GPA: 4.12/4.5

RESEARCH EXPERIENCE

Research Engineer | Robotics and Payload Control Systems Laboratory

LIG Defense & Aerospace

Jan. 2025 – Present

Pangyo, Republic of Korea

- **Guided Diffusion for Humanoid Whole-Body Control**
 - Implemented a cost-guided latent diffusion model for humanoid motion synthesis
 - Investigating conflicts between task guidance and dominant-mode bias when generating underrepresented motions
- **Hydraulic Full-Body Exoskeleton Control with Mission Payloads**
 - Developed gait-phase-dependent hydraulic force control for stable walking with mission payloads
 - Developed FFT-based discrimination between wearer-exoskeleton interaction forces and external impacts for active balance control
 - Built a GUI for real-time monitoring and high-rate data logging during controller validation

Graduate Researcher | Human-Robot Collaboration Research Laboratory

Hanyang University (Advisor: Prof. Wansoo Kim)

Mar. 2023 – Feb. 2025

Seoul, Republic of Korea

- **Whole-Body Control for Balancing a Quadruped Mobile Manipulator**
 - Designed and implemented a hierarchical whole-body controller with centroidal momentum regulation
 - Validated balance on a 15° slope under unknown object disturbances, where linear-only and no-momentum baselines failed
- **Learning-Based Whole-Body Manipulation with a Quadruped Mobile Manipulator**
 - Integrated an RL-based swing-up trajectory generator with whole-body tracking for a quadruped mobile manipulator
 - Identified feasibility gaps when applying low-dimensional RL-generated trajectories to whole-body robot control
- **Flexible Whole-Body Control Framework for Human-Robot Collaboration**
 - Developed a hierarchical whole-body controller with risk-driven continuous task-priority transitions
 - Validated tunable safety-comfort-efficiency trade-offs in blended coexistence and collaboration scenarios
- **Viewpoint-Aware VR Teleoperation with Force Feedback**
 - Developed viewpoint-aware pose and force transformations for intuitive VR teleoperation
 - Implemented virtual-robot inverse kinematics and haptic feedback through a physical leader manipulator

Undergraduate Project Assistant

Hanyang University ERICA

Mar. 2022 – Feb. 2023

Ansan, Republic of Korea

- **Multi-Robot Coordination for Office Assistance Services** (Advisor: Prof. Byung-Ju Yi)
 - Designed and implemented Firebase-based coordination of multiple Temi robots, including real-time communication and nearest-robot task allocation

Manuscript Under Review

- [1] J. P. Jang*, Y. S. Choi*, D. G. Lee, and W. S. Kim, “Towards Flexible Human-Robot Collaborations: A Dynamic Control Framework Bridging Coexistence and Collaboration,” manuscript under review. (*Equal contribution)

Peer-Reviewed Journal Articles

- [1] D.-G. Lee, H. J. Kim, Y. S. Choi, S. Ha, J. No, S. Hwang, J.-K. Ryu, and W. Kim, “Posture Stabilization Control Framework for Lower Extremity Exoskeletons Operation in Protective-mission-oriented Tasks,” *Journal of Institute of Control, Robotics and Systems*, vol. 32, no. 1, pp. 75–81, 2026.
- [2] Y. S. Choi, J. P. Jang, S. M. Ha, S. W. Hwang, and W. S. Kim, “Whole-Body Control for Balancing an Articulated Legged-Arm Robot with the Integration of Momentum Management and a Passively Connected Unknown Object,” *The Journal of Korea Robotics Society*, vol. 20, no. 1, pp. 112–119, 2025.

Conference Papers

- [1] J. P. Jang, S. H. Hwang, Y. S. Choi, S. W. Hwang, and W. S. Kim, “Reinforcement Learning-Based Whole-Body Control for Dynamic Manipulation of Passively Connected Objects with a Legged-Arm Robot,” *The Korea Robotics Society Annual Conference (KRoC)*, 2025.
- [2] Y. S. Choi, S. H. Hwang, and W. S. Kim, “Whole-Body Control of an Articulated Legged-Arm Robot for Maintaining Balance with a Passively Connected Unknown Object,” *The Korean Society for Precision Engineering (KSPE)*, 2024.
- [3] J. P. Jang, S. M. Choi, Y. S. Choi, J. S. Lee, S. W. Hwang, and W. S. Kim, “Teleoperation Control Framework for Enhancing Operator Telepresence through Real-Time Virtual Environment Integration,” *The Korea Robotics Society Annual Conference (KRoC)*, 2024.
- [4] Y. R. Yoon, S. H. Lee, Y. S. Choi, S. R. Do, J. C. Park, J. Y. Yang, J. H. Woo, and B. J. Yi, “Development of Office Assistance Services Using Mobile Robots,” *37th ICROS Annual Conference*, pp. 746–747, 2022.

HONORS & AWARDS

Undergraduate Student Paper Award , 37th ICROS Annual Conference	2022
University President’s Award (2nd Place) , Haedong Entrepreneurship Competition	2021
Selected Team , Student Entrepreneurship 300 Program, Ministry of Education	2021
Merit-Based Tuition Scholarship , Hanyang University ERICA	2017, 2020 – 2022

TEACHING EXPERIENCE

Teaching Assistant <i>Hanyang University ERICA</i>	Sep. 2023 – Dec. 2024 <i>Ansan, Republic of Korea</i>
- Developed laboratory materials and led hands-on sessions for a first-year robotics course - Designed the final project and evaluated student hardware demonstrations	

ENTREPRENEURSHIP & OUTREACH

Early Core Team Member DORO <i>AI and Robotics Education Startup</i>	Jul. 2020 – Feb. 2023 <i>Ansan, Republic of Korea</i>
- Co-developed and iteratively refined DORO’s early business model and educational strategy through pilot programs and service experimentation - Developed and delivered robot kits, instructional content, and educational programs, including an eight-part Arduino video series, community robotics workshops, and a line-tracing hackathon for students	

TECHNICAL SKILLS

Programming: C/C++, Python, C#, MATLAB
Frameworks & Tools: PyTorch, Isaac Lab, ROS/ROS 2, WPF, Linux, Git
Simulation: Isaac Sim, RaiSim, MuJoCo, Gazebo, Unity
Robot Platforms: Franka Research 3, UR5e, Robotnik SUMMIT-XL STEEL, Temi, custom full-body exoskeleton
Hardware & Sensors: qb SoftHand Research, ATI AXIA-80 F/T sensor, Xsens MVN Awinda, OptiTrack